Type MS · Unit vectors: same direction, length one

Answer 1 of 12 · Question 6 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	eg $\overrightarrow{SR} = \overrightarrow{SP} + \overrightarrow{PR} = (3\mathbf{i} - 15\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= 2\mathbf{i} + 3\mathbf{j}$ or eg $\overrightarrow{QR} = \overrightarrow{QP} + \overrightarrow{QR} = (-2\mathbf{i} - 3\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= -3\mathbf{i} + 15\mathbf{j}$	M1 A1 M1 A1
	eg $\overrightarrow{PQ} = \overrightarrow{SR}$ or $\overrightarrow{PS} = \overrightarrow{QR}$ oe	M1
	(As opposite sides are equal in length and parallel) $PQRS$ is a parallelogram *	A1cso [4]
(b)	$\overrightarrow{QS} = \overrightarrow{QP} + \overrightarrow{PS} = (-2\mathbf{i} - 3\mathbf{j} - 3\mathbf{i} + 15\mathbf{j})$ $= -5\mathbf{i} + 12\mathbf{j}$	M1 A1
	Unit vector = $(\pm)\frac{1}{13}(-5\mathbf{i} + 12\mathbf{j})$	M1 A1 [4]
(c)	$\overrightarrow{PT} = 2\mathbf{i} + 3\mathbf{j} + \frac{5}{13}("-5\mathbf{i} + 12\mathbf{j}") = \frac{1}{13}\mathbf{i} + \frac{99}{13}\mathbf{j}$	M1 A1 [2]
		Total 10 marks

Type MS · Unit vectors: same direction, length one

Answer 1 of 12 · Question 6 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 2 of 12 · Question 7 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

number		
7 (a)	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$ $\overrightarrow{BC} = -(15\mathbf{i} - 6\mathbf{j}) + 8\mathbf{i} + \mathbf{j} = -7\mathbf{i} + 7\mathbf{j}$	M1 A1 [2]
(b)	$\left\|\overrightarrow{BC}\right\| = \sqrt{98} = \left(7\sqrt{2}\right)$ Unit vector is $\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe	B1 B1 [2]
(c)	$\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{MC}$ are parallel oe (and have same point of origin (M)) hence they are collinear. **ALT 1** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $\Rightarrow \overrightarrow{NC} = -(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{NC}$ are parallel oe (and share the same point (N)) hence they are collinear. **ALT 2** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MC}$ and $\overrightarrow{NC}$ are parallel oe(and share the same point (C)) hence they are collinear.	B1 M1 M1 A1 [4] {B1} {M1} {M1} {A1} [4] {B1} {M1} {M1} {A1} [4]
		Total 8 marks

Type MS · Unit vectors: same direction, length one

Answer 2 of 12 · Question 7 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(a)	
M1	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$
A1	$\overrightarrow{BC} = -7\mathbf{i} + 7\mathbf{j}$
(b)	
B1	$\sqrt{98}$ oe
B1	$\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe
(c)	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ (may be implied by $\overrightarrow{MN}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{MC} = 4\overrightarrow{MN}$
ALT 1	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{MN}$ or $\overrightarrow{NC}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $-(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = 3\overrightarrow{MN}$
ALT 2	
B1	$\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{NC}$)
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = \frac{3}{4}\overrightarrow{MC}$
	For part c: Send any geometrical solutions to review

Type MS · Unit vectors: same direction, length one

Answer 2 of 12 · Question 7 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 3 of 12 · Question 1 part b · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

1(a)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}, = -3\mathbf{i} + 4\mathbf{j}$	M1,A1 (2)
(b)	$\left\|\overrightarrow{AB}\right\| = 5$	
	Unit vector $= \frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ or $-\frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ oe Accept column vectors	M1A1 (2)
		[4]

(a)	
M1	For $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ seen, or $\overrightarrow{AB} = (\mathbf{i} + 7\mathbf{j}) - (4\mathbf{i} + 3\mathbf{j})$ or equivalent in column form
A1	Correct simplified answer as shown or equivalent but NOT a column vector
(b)	
M1	Correct modulus of their $\overrightarrow{AB}$ **and** divide ± their $\overrightarrow{AB}$ by it Correct unit vector, as shown or equivalent inc column vector
A1	$\pm\frac{1}{5}(-3\mathbf{i} + 4\mathbf{j})$ scores M1A0 $\frac{1}{5} - 3\mathbf{i} + 4\mathbf{j}$ scores M1A0
NB:	If $\overrightarrow{BA}$ is found in (a) both (b) marks are still available

Type MS · Unit vectors: same direction, length one

Answer 3 of 12 · Question 1 part b · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{d4}$	

Type MS · Unit vectors: same direction, length one

Answer 4 of 12 · Question 1 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Unit vectors: same direction, length one

Answer 4 of 12 · Question 1 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Type MS · Unit vectors: same direction, length one

Answer 4 of 12 · Question 1 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

International GCSE Further Pure Mathematics – Paper 2R mark scheme

Question number	Scheme	Marks
1(a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$ or $6\mathbf{i} + 8\mathbf{j} = \overrightarrow{OB} - (3\mathbf{i} - 2\mathbf{j})$ oe $9\mathbf{i} + 6\mathbf{j}$	M1 A1 [2]
(b)	$\sqrt{6^2 + 8^2}$ *or* 10 (from Pythagorean triple)	B1 [1]
(c)	$(\pm)\frac{1}{"10"}(6\mathbf{i} + 8\mathbf{j})$	M1 A1 [2]
		Total 5 marks

Type MS · Unit vectors: same direction, length one

Answer 5 of 12 · Question 4 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4(a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{i} + 7\mathbf{j} + a\mathbf{i} + 16\mathbf{j} = (5+a)\mathbf{i} + 23\mathbf{j}$ $\left(5\sqrt{29}\right)^2 = (5+a)^2 + 23^2 \Rightarrow 5+a = \pm\sqrt{196} \Rightarrow a = 9, -19$	M1A1 M1A1 [4]
(b)	$\overrightarrow{AB} = "9"\mathbf{i} + 16\mathbf{j} \Rightarrow \left\|\overrightarrow{AB}\right\| = \sqrt{"9"^2 + 16^2} = \sqrt{337}$ Unit vector: $\frac{1}{\sqrt{"337"}}("9"\mathbf{i} + 16\mathbf{j})$ oe.	M1 A1 [2]
		Total 6 marks

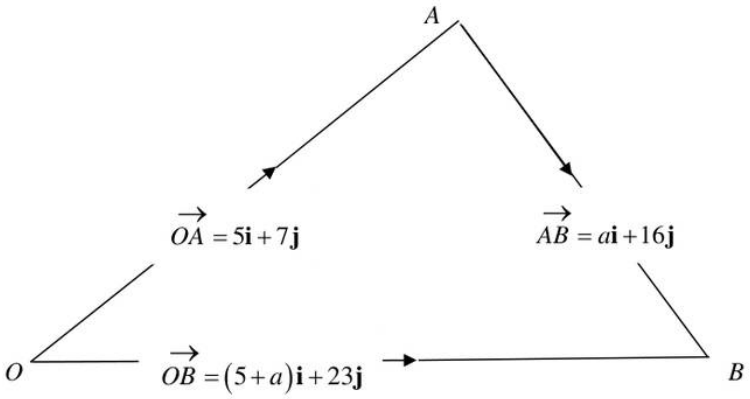
Part	Mark	Notes
(a)	M1	For the correct vector statement for $\overrightarrow{OB}$ For example, accept $\overrightarrow{AB} = \overrightarrow{A0} + \overrightarrow{OB}$ This mark can be implied by a correct vector for $\overrightarrow{OB}$
	A1	For the correct vector in terms of a [simplified or unsimplified].
	M1	For using Pythagoras theorem with $\left(5\sqrt{29}\right)$ and their vector for $\overrightarrow{OB}$ and solving the equation to find **two** values of a
	A1	For the **two** correct values. $a = 9, -19$ seen in their working
(b)	M1	For finding $\lvert AB\rvert$ by using a correct Pythagoras and writing down the unit vector where $\overrightarrow{AB} = k\mathbf{i} + 16\mathbf{j}$ where k is a **positive** value. Award for $\frac{"9"\mathbf{i} + 16\mathbf{j}}{\sqrt{"9"^2 + 16^2}}$ $\sqrt{9^2 + 16^2}$ can be implied by sight of $\sqrt{337}$ If they have an incorrect value for a, full substitution using Pythagoras theorem must be seen for the award of this mark.
	A1	For the correct unit vector. Accept any correct equivalent unit vectors. For example; $\frac{\sqrt{"337"}}{"337"}("9"\mathbf{i} + 16\mathbf{j})$ or $\frac{"9"\mathbf{i}}{\sqrt{"337"}} + \frac{16\mathbf{j}}{\sqrt{"337"}}$ Accept decimal answers. Eg., $0.054("9"\mathbf{i} + 16\mathbf{j})$ for awrt 0.054 Or $0.49\mathbf{i} + 0.87\mathbf{j}$ or better. Allow $\pm\frac{1}{\sqrt{"337"}}("9"\mathbf{i} + 16\mathbf{j})$

Type MS · Unit vectors: same direction, length one

Answer 5 of 12 · Question 4 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Useful Sketch



Type MS · Unit vectors: same direction, length one

Answer 5 of 12 · Question 4 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 6 of 12 · Question 4 part c · 1 mark

2020 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

4(a)	$\overrightarrow{DC} = (11\mathbf{i} - p\mathbf{j}) - (4\mathbf{i} - 2p\mathbf{j}) = 7\mathbf{i} + p\mathbf{j} = \overrightarrow{AB}$ OR: $\overrightarrow{BC} = (11\mathbf{i} - p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = 4\mathbf{i} - 2p\mathbf{j} = \overrightarrow{AD}$	M1A1	
	Parallel and equal in length ∴ Parallelogram	A1cso	(3)
(b)	$\overrightarrow{BD} = (4\mathbf{i} - 2p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = -3\mathbf{i} - 3p\mathbf{j}$ (or $3(-\mathbf{i} - p\mathbf{j})$ oe	B1	
	$\sqrt{9 + (3p)^2} = 3\sqrt{10}$ $(\Rightarrow 9 + 9p^2 = 90)$	M1	
	$p = \pm 3$	A1	(3)
(c)	$(\pm)\frac{1}{3\sqrt{10}}(-3\mathbf{i} - 9\mathbf{j})$ oe	B1ft	(1)
			[7]

	Accept column vectors throughout.
(a)	
M1	Attempt $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC}$ using the difference of 2 appropriate vectors in component form.
A1	Show that $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC} = \pm\underset{\sim}{AB}$ or $\pm\underset{\sim}{AD}$
A1cso	Suitable conclusion with reason from correct working. One pair of vectors only needed if reason is "parallel and equal". Both pairs needed if reason is "2 pairs of sides parallel/equal".
(b)	
B1	For a correct $\underset{\sim}{BD}$ or $\underset{\sim}{DB}$. No simplification needed.
M1	Use the given length of $\underset{\sim}{BD}$ with the length of their $\underset{\sim}{BD}$ to form an equation
A1	Obtain correct values for p. Both needed.
(c)	
B1ft	Use their positive value for p to obtain a unit vector (no simplification needed)

Type MS · Unit vectors: same direction, length one

Answer 6 of 12 · Question 4 part c · 1 mark

2020 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 7 of 12 · Question 3 part b · 5 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

3 a	$\left\|\overrightarrow{OA}\right\| = \sqrt{p^2+16}$ and $\left\|\overrightarrow{OB}\right\| = \sqrt{4p^2+4p+2}$	
	$\sqrt{2}\left\|\overrightarrow{OA}\right\| = \left\|\overrightarrow{OB}\right\| \Rightarrow 2p^2+32 = 4p^2+4p+2$	M1
	$2p^2+4p-30=0 \Rightarrow p^2+2p-15=0$	M1
	$(p+5)(p-3)=0$	M1
	$p=3$	A1 (4)
b	$\overrightarrow{AB} = -'3'\mathbf{i}+4\mathbf{j}+\mathbf{i}+(2\times'3'+1)\mathbf{j} = -2\mathbf{i}+11\mathbf{j}$	M1 A1
	$\left\|\overrightarrow{AB}\right\| = \sqrt{'4'+'121'}$ $\left[=5\sqrt{5}\right]$	M1
	$'\left[\frac{1}{5\sqrt{5}}\right]"(-2\mathbf{i}+11\mathbf{j})'$	dM1
	$(\pm)\ \frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})$	A1 (5)
		Total 9 marks

Type MS · Unit vectors: same direction, length one

Answer 7 of 12 · Question 3 part b · 5 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For use of $\sqrt{2}\left\|\overrightarrow{OA}\right\| = \left\|\overrightarrow{OB}\right\|$ i.e., $\sqrt{2}\times\sqrt{p^2+(-4)^2} = \sqrt{1+(2p+1)^2} \Rightarrow \left(\sqrt{2}\times\sqrt{p^2+16} = \sqrt{4p^2+4p+2}\right)$ They may find $\left\|\overrightarrow{OA}\right\|$ and $\left\|\overrightarrow{OB}\right\|$ separately. Award when combined with $\sqrt{2}$ and condone arithmetical slips.
	M1	For forming a 3TQ in any order. [The correct 3TQ is $2p^2+4p-30=0$ or $p^2+2p-15=0$]
	M1	For a correct attempt to solve **their** 3TQ by any valid method. They must reach a value of p for this mark.
	A1	For $p=3$ If they also give $p=-5$ without evidence of rejecting this solution, withhold the A mark
(b)	M1	For the vector statement $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$ o.e. This can be implied by sight of $\overrightarrow{AB} = \text{'}-2\text{'}\mathbf{i} + \text{'}11\text{'}\mathbf{j}$ or $\overrightarrow{AB} = \begin{pmatrix}\text{'}-2\text{'}\\ \text{'}11\text{'}\end{pmatrix}$ If there is no vector statement you must check their vector for substitution of their p.
	A1	For the correct $\overrightarrow{AB}$ (allow unsimplified) $\left(\overrightarrow{AB} = -2\mathbf{i}+11\mathbf{j}\right)$ and also allow $\overrightarrow{AB} = \begin{pmatrix}-2\\ 11\end{pmatrix}$ Award for sight of $-2\mathbf{i}+11\mathbf{j}$ only.
	M1	For using Pythagoras theorem on their $\overrightarrow{AB}$ i.e., $\sqrt{(\text{'}-2\text{'})^2+\text{'}11\text{'}^2}$
	dM1	For correct method to find a unit vector using their values and their $\overrightarrow{AB}$ NB: this mark is dependent on the previous M mark
	A1	For $\frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})$ oe $\left[\text{Allow } -\frac{1}{5\sqrt{5}}(-2\mathbf{i}+11\mathbf{j})\right]$ **OR** $\overrightarrow{AB} = \pm\frac{1}{5\sqrt{5}}\begin{pmatrix}-2\\ 11\end{pmatrix}$

Type MS · Unit vectors: same direction, length one

Answer 7 of 12 · Question 3 part b · 5 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$O_1A = O_2A = O_1O_2 = 6$ so triangle AO_1O_2 is equilateral	B1

Type MS · Unit vectors: same direction, length one

Answer 8 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3(a)	$\overrightarrow{OA} = \overrightarrow{OB} - \overrightarrow{AB} = 7i + 2aj - (i + 3aj) = 6i - aj$ $\left\|\overrightarrow{OA}\right\| = \sqrt{6^2 + a^2} = 3\sqrt{5} \Rightarrow 6^2 + a^2 = 45 \Rightarrow a^2 = 9 \Rightarrow a = 3$	M1A1 M1A1 [4]
(b)	$\overrightarrow{OA} = 6i - '3'\, j$ {Unit vector parallel to $\overrightarrow{OA}$ is} $\{\pm\}\frac{1}{3\sqrt{5}}(6i - 3j)$ oe ,isw	B1ft B1 [2]
		Total 6 marks

Part	Mark	Notes
(a)	M1	For the correct vector statement with vector $\overrightarrow{OA}$ or $\overrightarrow{AO}$ e.g. $\overrightarrow{OA} = \overrightarrow{OB} - \overrightarrow{AB}$ or $\left\{\overrightarrow{OA} =\right\} 7i + 2aj - (i + 3aj)$ oe
	A1	For the correct simplified or unsimplified vector $\overrightarrow{OA}$ **or** $\overrightarrow{AO}$
	M1	For using the correct Pythagoras theorem on their $\overrightarrow{OA}$ or $\overrightarrow{AO}$ with $3\sqrt{5}$, leading to a value for a^2 or a
	A1	For the correct value of a=3 only (must reject a=−3 if present) **NB:A value of a=3 must come from a correct vector of OA with no incorrect work seen.**
(b)	B1ft	For the correct vector for $\overrightarrow{OA}$ or $\overrightarrow{AO}$ using their a (which a>0), (Substitutes their a value to $\overrightarrow{OA} = 6i - aj$ or $\overrightarrow{AO} = -6i + aj$
	B1	Correct unit vector $\frac{1}{3\sqrt{5}}(6i - 3j)$ *oe*, can be in either direction, so accept + or − isw

Extra notes:

1. Allow working in column vectors throughout.
2. Condone missing arrow on their vectors.
3. **Special Case for part a:** Correct answer with minimum work, no incorrect working shown
e.g. $\sqrt{6^2 + a^2} = 3\sqrt{5}$ M1A1M1 $\rightarrow 6^2 + a^2 = 45 \rightarrow a^2 = 9 \rightarrow a = 3$ A1

Type MS · Unit vectors: same direction, length one

Answer 8 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 9 of 12 · Question 10 part whole · 10 marks

2023 JUN · Paper 01 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Unit vectors: same direction, length one

Answer 10 of 12 · Question 4 part c · 4 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
4(a)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} \Rightarrow \overrightarrow{AB} = (5\mathbf{i} + 9p\mathbf{j}) - (p\mathbf{i} + 2p\mathbf{j})$	M1A1
	$\mathbf{i}(5-p) + \mathbf{j}(7p) = Q(\mathbf{i} - 2\mathbf{j}) \Rightarrow 5 - p = Q$ and $7p = -2Q$	M1M1
	$7p = -2(5-p) \Rightarrow p = -2$	M1A1 [6]
(b)	$7('-2') = -2Q \Rightarrow Q = 7$, $\overrightarrow{AB} = 7(\mathbf{i} - 2\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	M1A1ft [2]
	OR	
	$\overrightarrow{AB} = (5\mathbf{i} + 9(-2)\mathbf{j}) - ((-2)\mathbf{i} + 2(-2)\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	[M1A1ft]
(c)	$\overrightarrow{OA} = -2\mathbf{i} - 4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2 + (-4)^2} = \sqrt{20}$	M1A1ft
	Unit vector is $\frac{1}{\sqrt{20}}(-2\mathbf{i} - 4\mathbf{j}) = \frac{\sqrt{5}}{5}(-\mathbf{i} - 2\mathbf{j})$	M1A1 [4]
		Total 12 marks

Type MS · Unit vectors: same direction, length one

Answer 10 of 12 · Question 4 part c · 4 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Question	Notes	Marks
4(a)	For the basic vector statement $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$	M1
	For the correct vector (simplified or unsimplified) $\overrightarrow{AB} = (5\mathbf{i}+9p\mathbf{j}) - (p\mathbf{i}+2p\mathbf{j}) = [\mathbf{i}(5-p)+\mathbf{j}(7p)]$	A1
	For setting their $\overrightarrow{AB} = Q(\mathbf{i}-2\mathbf{j})$ where $Q \neq 1,\ Q \neq 0$ $\mathbf{i}(5-p)+\mathbf{j}(7p) = Q(\mathbf{i}-2\mathbf{j})$	M1
	For equating components of **i** and **j** **i** $\quad 5-p = Q$ **j** $\quad 7p = -2Q$	M1
	Solving the simultaneous equations by any method to find the value of p $7p = -2(5-p) \Rightarrow p = ...$	M1
	For the value of $p = -2$	A1 [6]
(b)	For finding the value of k and using it to find the vector $\overrightarrow{AB}$ $7('-2') = -2Q \Rightarrow Q = 7$	M1
	For the correct vector $\overrightarrow{AB} = 7(\mathbf{i}-2\mathbf{j}) = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
	ALT	
	For substituting their value of p to find the vector $\overrightarrow{AB}$ $\overrightarrow{AB} = \mathbf{i}(5-[-2])+\mathbf{j}(7[-2]) = ...$	M1
	$\overrightarrow{AB} = \mathbf{i}(5-p)+\mathbf{j}(7p) \Rightarrow \overrightarrow{AB} = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
(c)	$\overrightarrow{OA} = -2\mathbf{i}-4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2+(-4)^2} = ...$	M1
	$\left\lvert\overrightarrow{OA}\right\rvert = \sqrt{20}$	A1ft
	Unit vector in the direction of $\overrightarrow{OA}$ is $\frac{1}{\sqrt{20}}(-2\mathbf{i}-4\mathbf{j})$	M1
	Unit vector in the required form $\frac{\sqrt{5}}{5}(-\mathbf{i}-2\mathbf{j})$ Allow $\frac{\sqrt{5}}{5}(\mathbf{i}+2\mathbf{j})$ **provided** no processing errors seen.	A1 [4]
		Total 12 marks

Type MS · Unit vectors: same direction, length one

Answer 10 of 12 · Question 4 part c · 4 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 11 of 12 · Question 6 part b · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Number		
6 (a)	$\left(\overrightarrow{AB}\right) = \overrightarrow{AO} + \overrightarrow{OB} = -9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$	M1
	$(-7)^2 + (p-12)^2 = 25^2$	M1
	$(p-12)^2 = 576$	
	$p = 12 \pm \sqrt{576}$	M1
	$p = -12, 36^*$	A1*
		[4]
(b)	$\sqrt{12^2 + 2^2} \Rightarrow \left(\sqrt{148} = 2\sqrt{37}\right)$	M1
	$\pm\frac{1}{\sqrt{37}}(\mathbf{a} - 6\mathbf{b})$ **or** $\pm\frac{\sqrt{37}}{37}(\mathbf{a} - 6\mathbf{b})$	A1
		[2]
(c)	$(\overrightarrow{OC}) = \frac{2}{5}(2\mathbf{a} + 36\mathbf{b})$	M1
	$(\overrightarrow{CA}) = -\frac{2}{5}(2\mathbf{a} + 36\mathbf{b}) + 9\mathbf{a} + 12\mathbf{b}$	M1
	$= \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$	A1
		[3]
		Total 9 marks

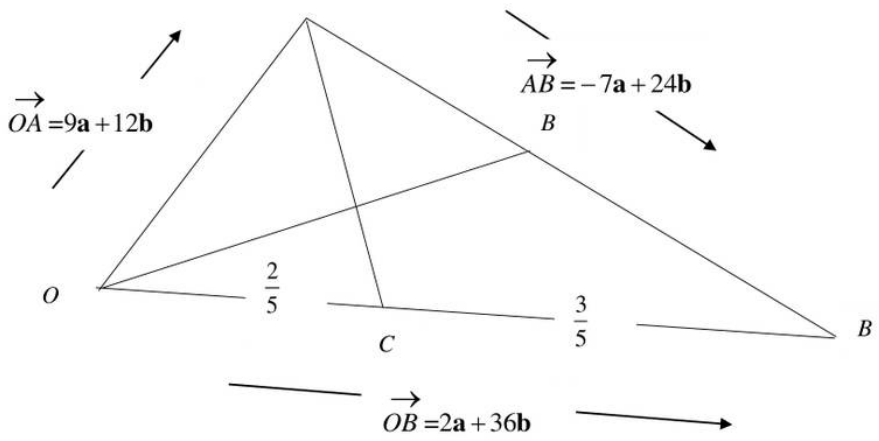
Type MS · Unit vectors: same direction, length one

Answer 11 of 12 · Question 6 part b · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
6 (a)	M1	For a correct vector path to find $\pm\overrightarrow{AB}$ or $-9\mathbf{a} - 12\mathbf{b} + 2\mathbf{a} + p\mathbf{b}$ or $9\mathbf{a} + 12\mathbf{b} - 2\mathbf{a} - p\mathbf{b}$
	M1	For using Pythagoras' Theorem with th B $\overrightarrow{AB}$
	M1	For any complete and minimally acceptable attempt to solve for p See General Guidance.
	A1*	For both correct values of p – no errors seen. **Note:** This is a show question and the answers are given
(b)	M1	For a corr od to find the magnitude.
O	A1	For a corr $\frac{2}{5}$ vector as shown.
(c)	M1	For a corr or fc $\overrightarrow{C}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CO} + \overrightarrow{OA}$
	A1	For $\frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$
	ALT: Uses the vector path $\overrightarrow{CA} = \overrightarrow{CB} + \overrightarrow{BA}$	
	M1	For a correct vector for $\overrightarrow{CB} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b})$ or $\overrightarrow{BA} = 7\mathbf{a} - 24\mathbf{b}$
	M1	For the correct vector for $\overrightarrow{CA}$ using their $\overrightarrow{CB}$ and $\overrightarrow{BA}$ $\overrightarrow{CA} = \frac{3}{5}(2\mathbf{a} + 36\mathbf{b}) + (7\mathbf{a} - 24\mathbf{b})$
	A1	$\overrightarrow{CA} = \frac{41}{5}\mathbf{a} - \frac{12}{5}\mathbf{b}$

Useful Sketch



Type MS · Unit vectors: same direction, length one

Answer 11 of 12 · Question 6 part b · 2 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Unit vectors: same direction, length one

Answer 12 of 12 · Question 9 part b · 4 marks

2025 NOV · Paper 02 · extract 1 of 5 · find part b in this mark scheme extract

	Scheme	Marks
9(a)	$\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$ or $-(2\mathbf{a}+4\mathbf{b})+(7\mathbf{a}+7\mathbf{b})$	B1
	$= 5\mathbf{a}+3\mathbf{b}$	B1
	$\overrightarrow{BC}\left[= \frac{1}{2}(10\mathbf{a}+6\mathbf{b})=\right] \frac{1}{2}\overrightarrow{AD}$ oe eg $\frac{1}{2}(10\mathbf{a}+6\mathbf{b}) = 5\mathbf{a}+3\mathbf{b}$ eg $10\mathbf{a}+6\mathbf{b} = 2(5\mathbf{a}+3\mathbf{b})$	M1
	[therefore parallel] therefore trapezium or eg # or shown.	A1 [4]
(b)	$\left[\overrightarrow{BD} = -\overrightarrow{AB} + \overrightarrow{AD} = -(2\mathbf{a}+4\mathbf{b})+(10\mathbf{a}+6\mathbf{b}) =\right] 8\mathbf{a}+2\mathbf{b}$	B1
	$\left[\lvert BD\rvert =\right] \sqrt{8^2+2^2}\left[=\sqrt{68}=2\sqrt{17}\right]$	B1ft
	Unit vector is $\pm\frac{1}{\sqrt{68}}(8\mathbf{a}+2\mathbf{b})$ oe eg $\pm\frac{1}{\sqrt{17}}(4\mathbf{a}+\mathbf{b})$	M1A1 [4]

Type MS · Unit vectors: same direction, length one

Answer 12 of 12 · Question 9 part b · 4 marks

2025 NOV · Paper 02 · extract 2 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

Type MS · Unit vectors: same direction, length one

Answer 12 of 12 · Question 9 part b · 4 marks

2025 NOV · Paper 02 · extract 3 of 5 · find part b in this mark scheme extract

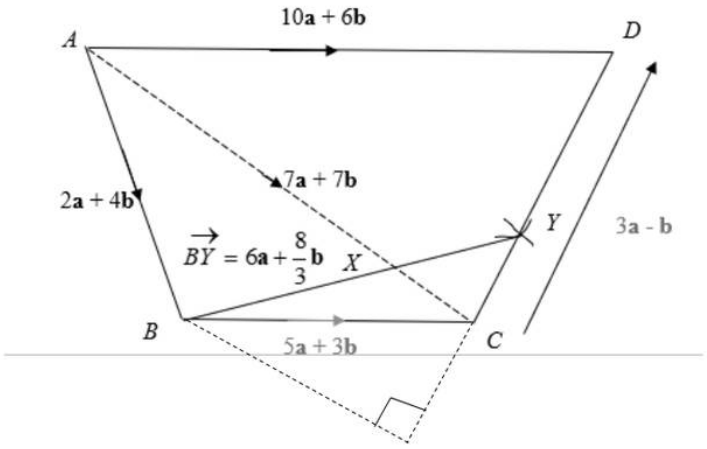
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$ $\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[6\lambda=7\mu-2$ and $\frac{8}{3}\lambda=7\mu-4$ or $\frac{8}{3}\lambda=-7\alpha+3$ and $6\lambda=-7\alpha+5\right]$ $\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$ $\left[6\beta=7\gamma+1$ and $\frac{8}{3}\beta=7\gamma-\frac{1}{3}$ or $\frac{8}{3}\beta=-7\delta+6+\frac{2}{3}$ and $6\beta=-7\delta+10-2\right]$ $\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]

Type MS · Unit vectors: same direction, length one

Answer 12 of 12 · Question 9 part b · 4 marks

2025 NOV · Paper 02 · extract 4 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



Type MS · Unit vectors: same direction, length one

Answer 12 of 12 · Question 9 part b · 4 marks

2025 NOV · Paper 02 · extract 5 of 5 · find part b in this mark scheme extract

(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\text{ and }\frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks